

The Binomial Distribution

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Introduction

The binomial distribution counts the number of successes in n independent trials, each with the same probability p . It is the most common discrete distribution in applied statistics: responder counts in a trial, positive test results in a screen, correct classifications in an evaluation.

Prerequisites

Bernoulli trials, basic combinatorics.

Theory

If $X = \sum_{i=1}^n B_i$ where B_i are iid Bernoulli(p), then $X \sim \text{Binomial}(n, p)$.

PMF: $P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$ for $k = 0, 1, \dots, n$.

Moments:

- $E[X] = np$.
- $\text{Var}(X) = np(1-p)$.

Normal approximation: $\text{Binomial}(n, p) \approx \mathcal{N}(np, np(1-p))$ when $np \geq 10$ and $n(1-p) \geq 10$. Continuity-corrected: $P(X \leq k) \approx \Phi((k + 0.5 - np) / \sqrt{np(1-p)})$.

Poisson approximation: for large n and small p with $np = \lambda$ moderate, $\text{Binomial}(n, p) \approx \text{Poisson}(\lambda)$.

Relationship to other distributions:

- Sum of independent binomials with same p : binomial with sum of n 's.
- Conditional on total successes, two independent binomials have a hypergeometric distribution of one.

Assumptions

- Fixed number of trials n .
- Independent trials.
- Constant success probability p .

Violations produce overdispersion (for varying p) or correlation (for dependent trials).

R Implementation

```
library(ggplot2)

n <- 20; p <- 0.35

k <- 0:n
pmf <- dbinom(k, n, p)

data.frame(k, pmf = round(pmf, 4))
```

```

# Moments
c(mean = n * p, var = n * p * (1 - p))
mean(rbinom(1e5, n, p)); var(rbinom(1e5, n, p))

# Normal approximation
x <- seq(0, n, length.out = 400)
norm_approx <- dnorm(x, mean = n * p, sd = sqrt(n * p * (1 - p)))

ggplot() +
  geom_col(data = data.frame(k, pmf), aes(k, pmf), fill = "#2A9D8F") +
  geom_line(data = data.frame(x, norm_approx), aes(x, norm_approx),
            colour = "#F4A261", linewidth = 1) +
  labs(x = "k", y = "probability",
       title = "Binomial(20, 0.35) with normal approximation") +
  theme_minimal()

# Interval probability and quantile
pbinom(10, n, p)      # P(X <= 10)
qbinom(0.025, n, p)   # 2.5th percentile

```

Output & Results

```

      k    pmf
1     0 0.0020
2     1 0.0219
...
20    20 0.0000

```

```

      mean    var
      7.0     4.55

```

```

[1] 0.9468      # P(X <= 10)
[1]      3      # 2.5th percentile

```

The normal approximation tracks the PMF closely at $n = 20$, $p = 0.35$ where $np = 7$ and $n(1 - p) = 13$.

Interpretation

Binomial reasoning pervades applied statistics: one-proportion tests, two-proportion tests, logistic regression, and every chi-squared-style comparison. Reporting a proportion with a Wilson 95% CI is reporting a binomial inference.

Practical Tips

- For small n or extreme p , use the exact binomial test / Clopper-Pearson interval instead of normal-based inference.
- Overdispersion relative to binomial signals either varying p across trials or correlation among trials; the beta-binomial is a common remedy.
- `dbinom(k, n, p, log = TRUE)` gives log-probabilities, useful for likelihoods with many observations.
- Compound inferences (e.g., comparing two binomial proportions) have their own CIs; prefer score intervals (`prop.test`) over Wald.
- In R the order of arguments is `dbinom(x, size, prob)` where `size` is n and `prob` is p ; mixing them up is a common bug.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren't.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Probability, conditional probability, Bayes' theorem
- Discrete distributions: Bernoulli, binomial, Poisson
- Continuous distributions and Q-Q plots